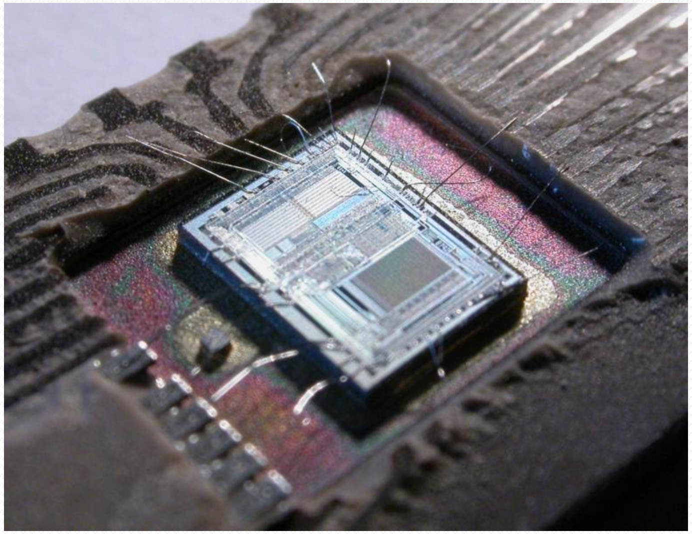
- A **microcontroller** (sometimes abbreviated **µC**, **uC** or **MCU**) is a small computer on a single **integrated circuit** containing a **processor core**, **memory**, and programmable **input/output** peripherals.
- It can only perform simple/specific tasks.
- A microcontroller is often described as a '**computer-on-a-chip**'.

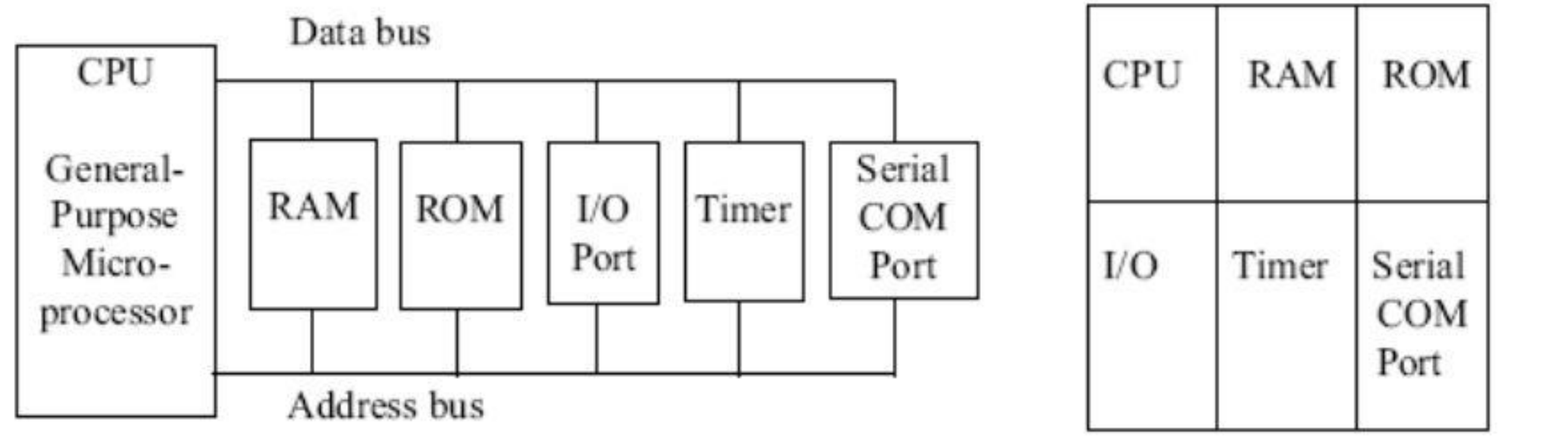


Definition of microcontroller

- A microcontroller is a computer present in a single integrated circuit which is dedicated to perform one task and execute one specific application.
- It contains memory, programmable input/output peripherals as well a processor. Microcontrollers are mostly designed for embedded applications and are heavily used in automatically controlled electronic devices such as cellphones, cameras, microwave ovens, washing machines, etc.

Microcontroller vs. General-Purpose Microprocessor (cont.)

- Timer
- ADC and other peripherals



(a) General-Purpose Microprocessor System
(b) Microcontroller

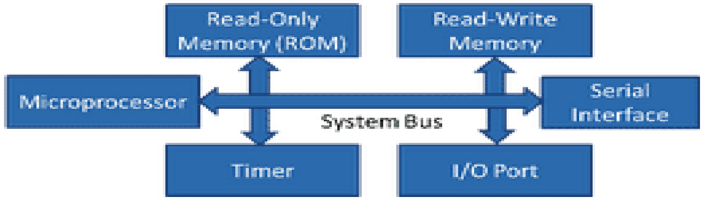
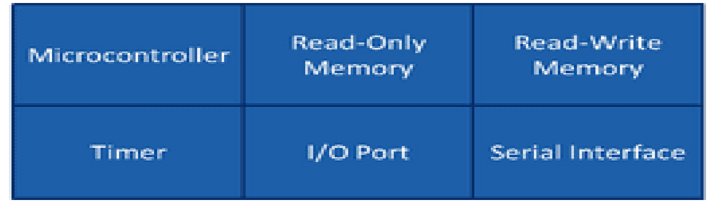
Microprocessor Vs Microcontroller

Microprocessor

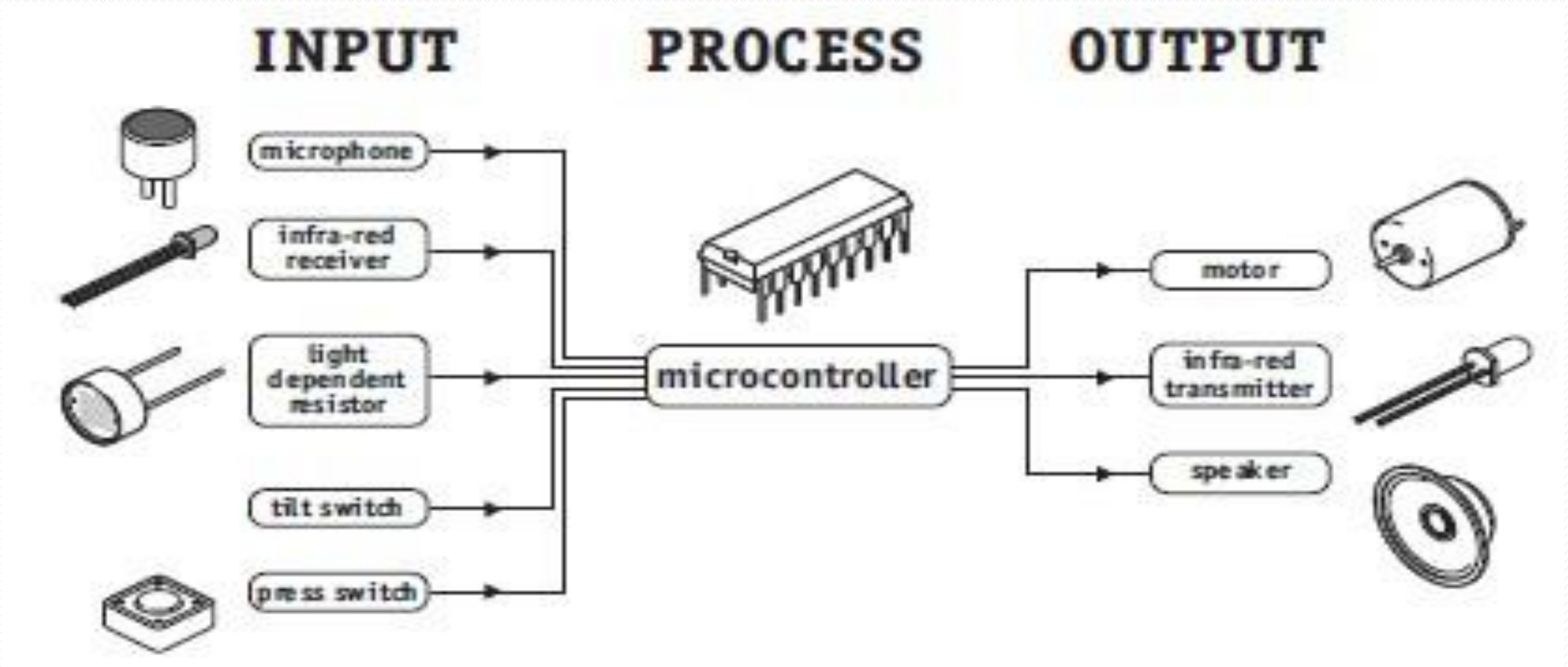
- Collection of on/off SWI on Silicon for computations
- CPU is stand-alone, RAM, ROM, I/O, timer are separate
- designer can decide on the amount of ROM, RAM and I/O ports.
- Expandable
- Expensive
- High Speed (1000 MIPS)
- General purpose and Versatile
- Large Architecture (32b, 64b)
- Lots of IO and Peripherals externally connected

Microcontroller

- Small and self-suffice SoC to control devices
- CPU, RAM, ROM, I/O and timer are all on a single chip
- fix amount of on-chip ROM, RAM, I/O ports
- Not Expandable – no external bus interface
- for applications in which cost, power and space are critical
- Low speed, on the order of 10 KHz – 20 MHz
- single-purpose
- Small usually an 8b
- Limited IO, enough for intended app

Microprocessor	Micro Controller
	
Microprocessor is heart of Computer system.	Micro Controller is a heart of embedded system.
It is just a processor. Memory and I/O components have to be connected externally	Micro controller has external processor along with internal memory and i/O components
Since memory and I/O has to be connected externally, the circuit becomes large.	Since memory and I/O are present internally, the circuit is small.
Cannot be used in compact systems and hence inefficient	Can be used in compact systems and hence it is an efficient technique
Cost of the entire system increases	Cost of the entire system is low
Due to external components, the entire power consumption is high. Hence it is not suitable to used with devices running on stored power like batteries.	Since external components are low, total power consumption is less and can be used with devices running on stored power like batteries.
Most of the microprocessors do not have power saving features.	Most of the micro controllers have power saving modes like idle mode and power saving mode. This helps to reduce power consumption even further.
Since memory and I/O components are all external, each instruction will need external operation, hence it is relatively slower.	Since components are internal, most of the operations are internal instruction, hence speed is fast.
Microprocessor have less number of registers, hence more operations are memory based.	Micro controller have more number of registers, hence the programs are easier to write.
Microprocessors are based on von Neumann model/architecture where program and data are stored in same memory module	Micro controllers are based on Harvard architecture where program memory and Data memory are separate
Mainly used in personal computers	Used mainly in washing machine, MP3 players

- Microcontrollers are purchased ‘blank’ and then programmed with a specific control program.
- Once programmed the microcontroller is build into a product to make the product more intelligent and easier to use.
- A designer will use a Microcontroller to:
  - Gather input from various sensors
  - Process this input into a set of actions
  - Use the output mechanisms on the microcontroller to do something useful.



PIC Microcontrollers

- Peripheral Interface Controller (PIC) was originally designed by General Instruments
- In the late 1970s, GI introduced PIC® 1650 and 1655 – RISC with 30 instructions.
- PIC was sold to Microchip
- Features: low-cost, self-contained, 8-bit, Harvard structure, pipelined, RISC, single accumulator, with fixed reset and interrupt vectors.

PIC Microcontroller product family

- 8-bit microcontrollers
  - PIC10
  - PIC12
  - PIC14
  - PIC16
  - PIC17
  - PIC18
- 16-bit microcontrollers
  - PIC24F
  - PIC24H
- 32-bit microcontrollers
  - PIC32
- 16-bit digital signal controllers
  - dsPIC30
  - dsPIC33F

PIC Microcontroller product family..

- The **F** in a name generally indicates the PICmicro uses flash memory and can be erased electronically.
- The **C** generally means it can only be erased by exposing the die to ultraviolet light (which is only possible if a windowed package style is used). An exception to this rule is the PIC16C84 which uses EEPROM and is therefore electrically erasable.

Table 1.3 Some PIC16CXXX and PIC16FXXX family members

Microcontroller	Program memory	Data RAM	Max speed (MHz)	I/O ports	A/D converter
PIC17C42	2048 × 16	232	33	33	–
16C554	512 × 14	80	20	13	–
16C64	2048 × 14	128	20	33	–
16C71	1024 × 14	68	20	13	4
16F877	8192 × 14	368	20	33	8
16F84	1024 × 14	68	10	13	–

An Example: PIC16F877

- **Why PIC16F877A is very popular?**
- This is because PIC16F877A is very cheap. Apart from that it is also very easy to be assembled. Additional components that you need to make this IC work is just a 5V power supply adapter, a 20MHz crystal oscillator and 2 units of 22pF capacitors.
- **What is the advantages of PIC16F877A?**
- This IC can be reprogrammed and erased up to 10,000 times.
- Therefore it is very good for new product development phase.
- **What is the disadvantages of PIC16F877A?**
- This IC has no internal oscillator so you will need an external crystal of other clock source.

Features

Key Features	PIC16F877
MAX Operating Frequency	20MHz
FLASH Program Memory (14-bit words)	8K
Data Memory (bytes)	368
EEPROM Data Memory (bytes)	256
I/O Ports	RA0-5 (6) RB0-7 (8) RC0-7 (8) RD0-7 (8) RE0-2 (3)
Timers	3
CCP (Capture/Compare/PWM)	2
Serial Communications	MSSP, USART
Parallel Communications	PSP
10-bit Analog-to-Digital Module	8 Channels
Instruction Set	35 Instructions
Pins (DIP)	40 Pins

Master Synchronous Serial Port

Parallel Slave Port (PSP)

PIC16F887 Basic Features

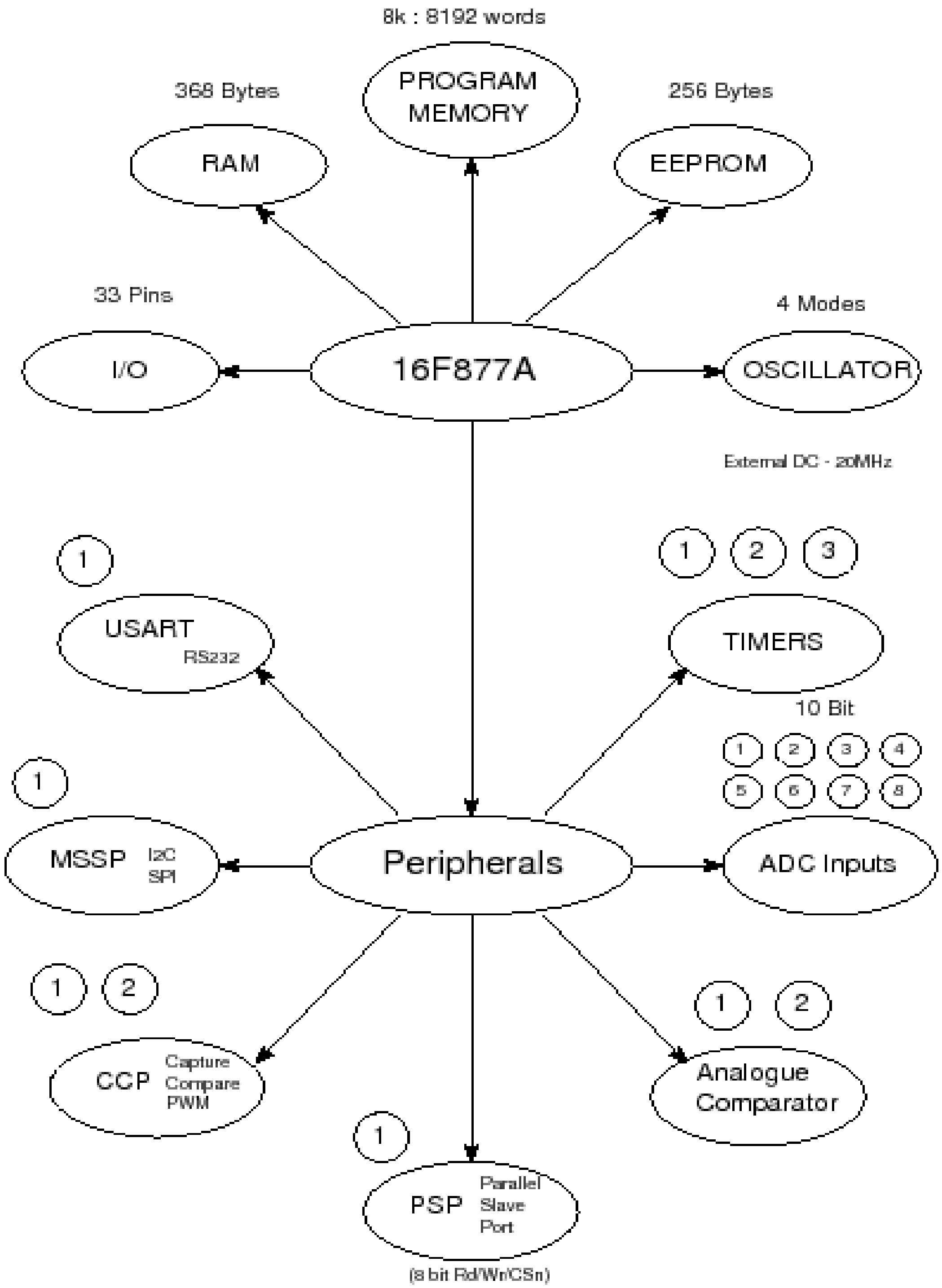
RISC architecture - Only 35 instructions to learn
All single-cycle instructions except branches
Operating frequency 0-20 MHz
Precision internal oscillator - Factory calibrated, Software selectable frequency range of 8MHz to 31KHz
Power supply voltage 2.0-5.5V - Consumption: 220uA (2.0V, 4MHz), 11uA (2.0 V, 32 KHz) 50nA (stand-by mode)
Power-Saving Sleep Mode
Brown-out Reset (BOR) with software control option
35 input/output pins
High current source/sink for direct LED drive
software and individually programmable pull-up resistor
Interrupt-on-Change pin
8K ROM memory in FLASH technology -Chip can be reprogrammed up to 100.000 times
In-Circuit Serial Programming Option -Chip can be programmed even embedded in the target device
256 bytes EEPROM memory - Data can be written more than 1.000.000 times
368 bytes RAM memory

https://www.ablic.com/en/semicon/products/automotive/automotive-watchdog-tim

A/D converter: 14-channels, 10-bit resolution
3 independent timers/counters
Watch-dog timer
Analogue comparator module with
Two analogue comparators
Fixed voltage reference (0.6V)
Programmable on-chip voltage reference
PWM output steering control
Enhanced USART module
Supports RS-485, RS-232 and LIN2.0
Auto-Baud Detect
Master Synchronous Serial Port (MSSP)
supports SPI and I2C mode

https://www.mikroe.com/ebooks/pic-microcontrollers-programming-in-c/pic16f887-basic-features

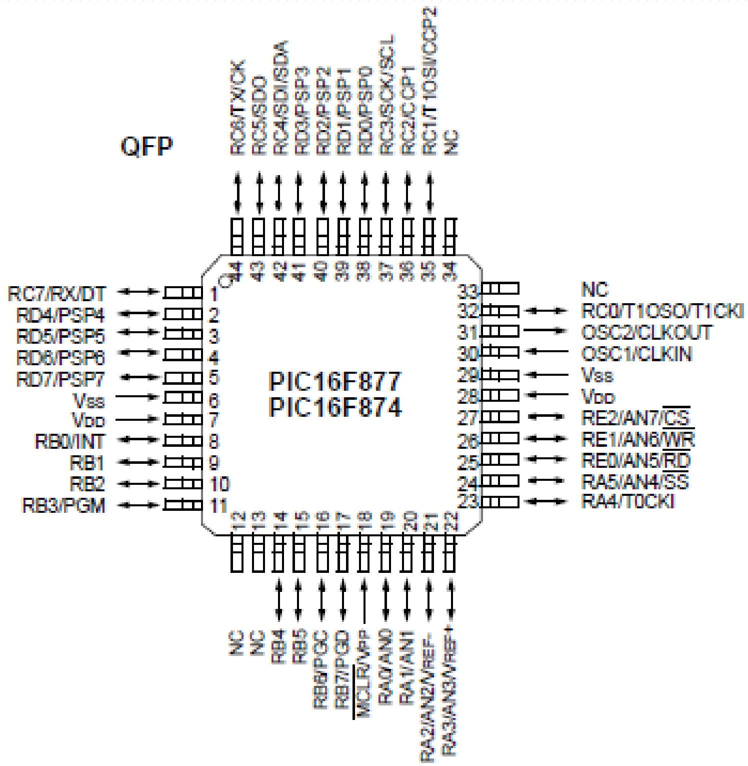
Bubble diagram of PIC16F877

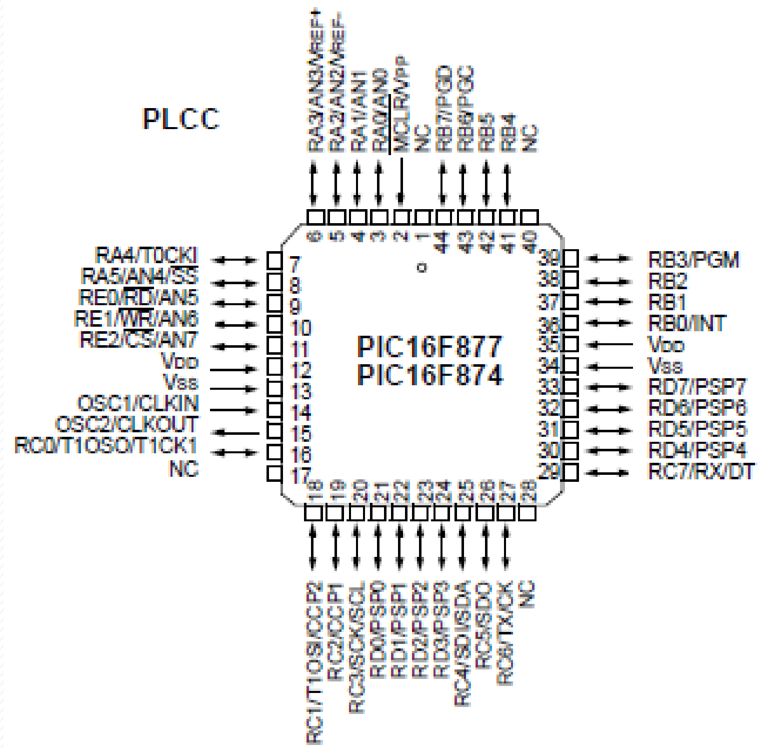


Pin Diagram of PIC16F877

- **Quad Flat Package (QFP)**

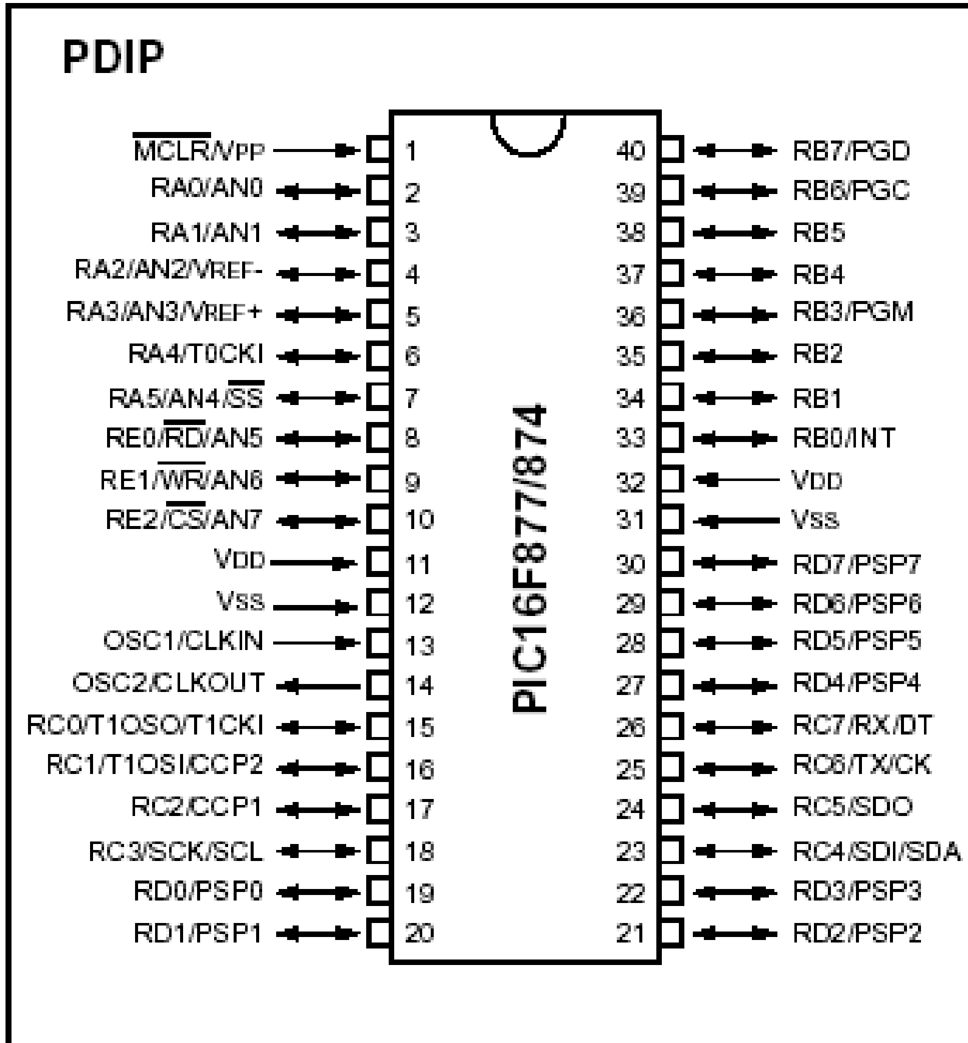
- **Plastic Leaded Chip Carrier Package (PLCC)**





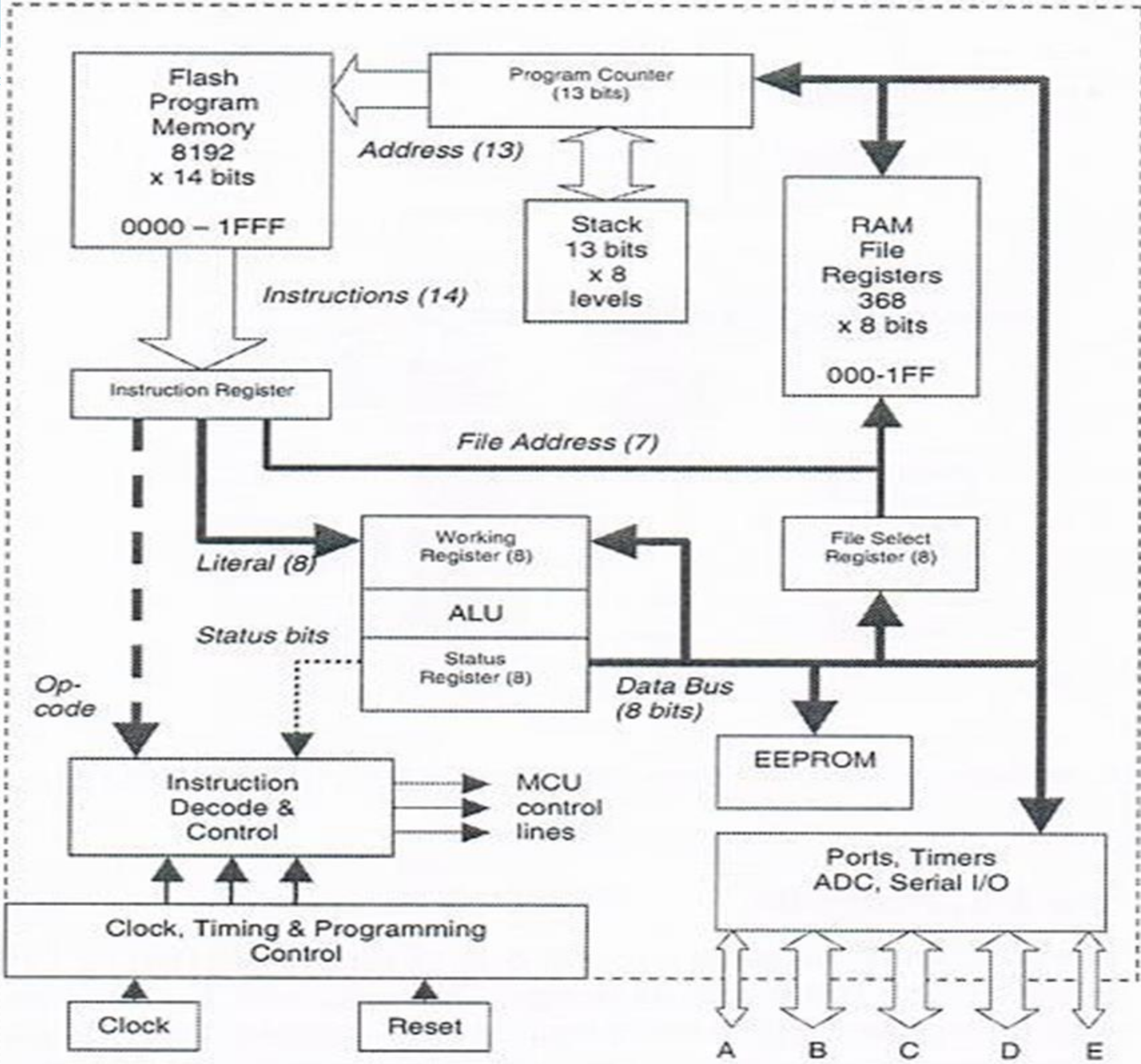
Pin Diagram of PIC16F877..

- **Plastic dual in-line package (DIP)**



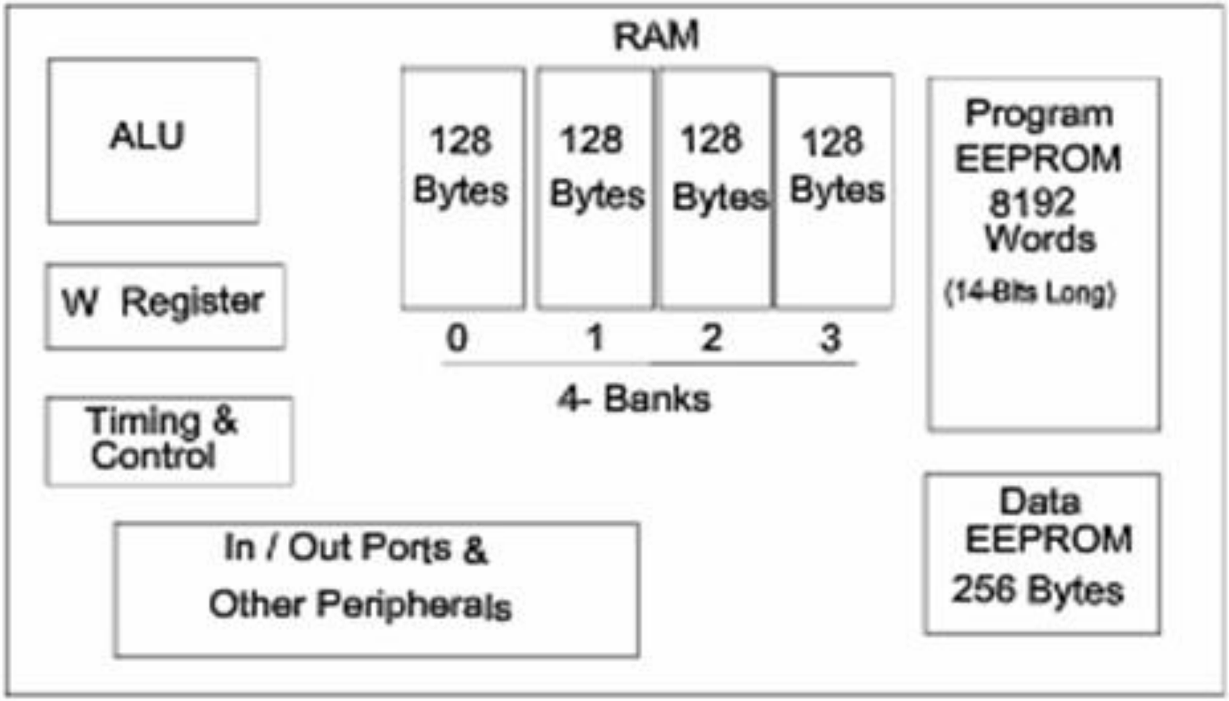
PIC16F877 Architecture

8-bit-The PIC16F877A-I/P is a 8-bit CMOS Flash-based Microcontroller



PIC16F877 Internal Block Diagram

- The basic architecture of PIC16F877 consists of Program memory, file registers and RAM, ALU and CPU registers.



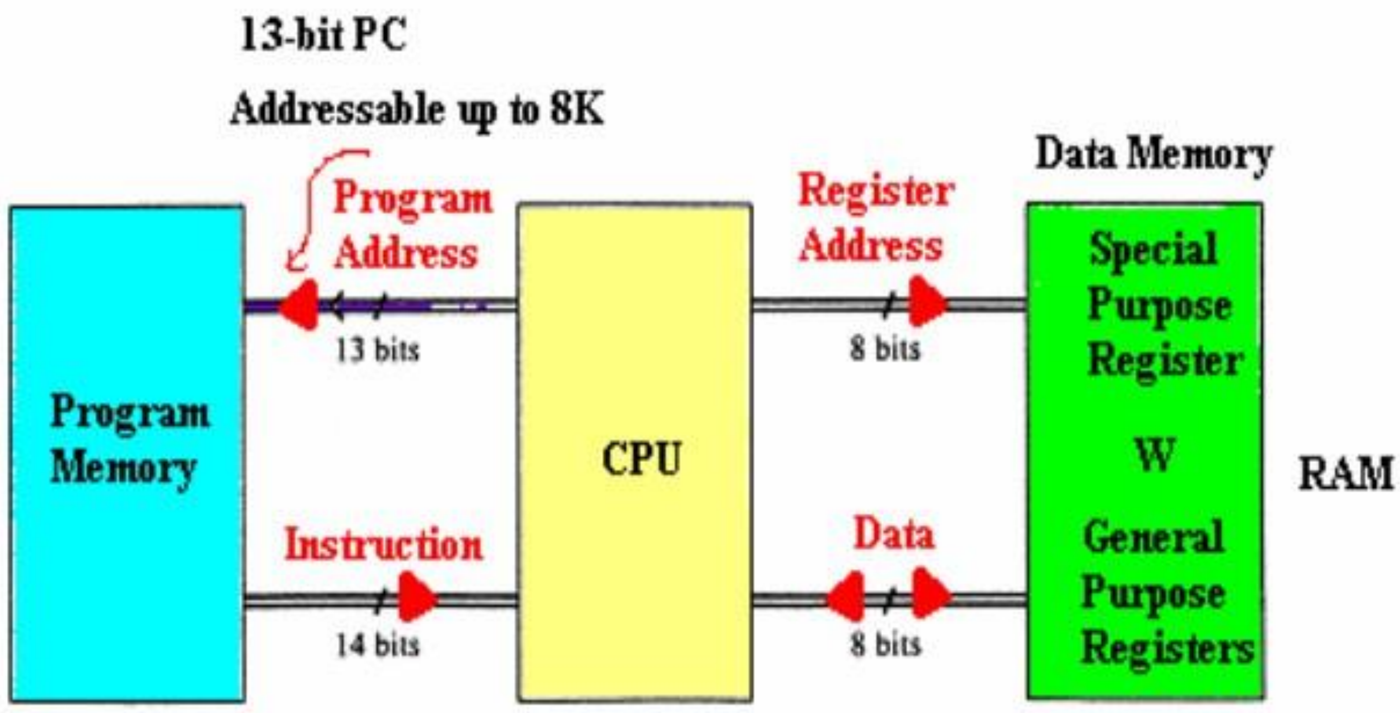
Memory of the PIC16F877

- divided into 3 types of memories:

1. **Program Memory** – A memory that contains the program (which we had written), after we've burned it. As a reminder, Program Counter executes commands stored in the program memory, one after the other.
2. **Data Memory** – This is RAM memory type, which contains a special registers like **SFR** (Special Function Register) and **GPR** (General Purpose Register). The variables that we store in the Data Memory during the program are deleted after we turn off the micro. These two memories have separated data buses, which makes the access to each one of them very easy.
3. **Data EEPROM** **(Electrically Erasable Programmable Read-Only Memory)** – A memory that allows storing the variables as a result of burning the written program.

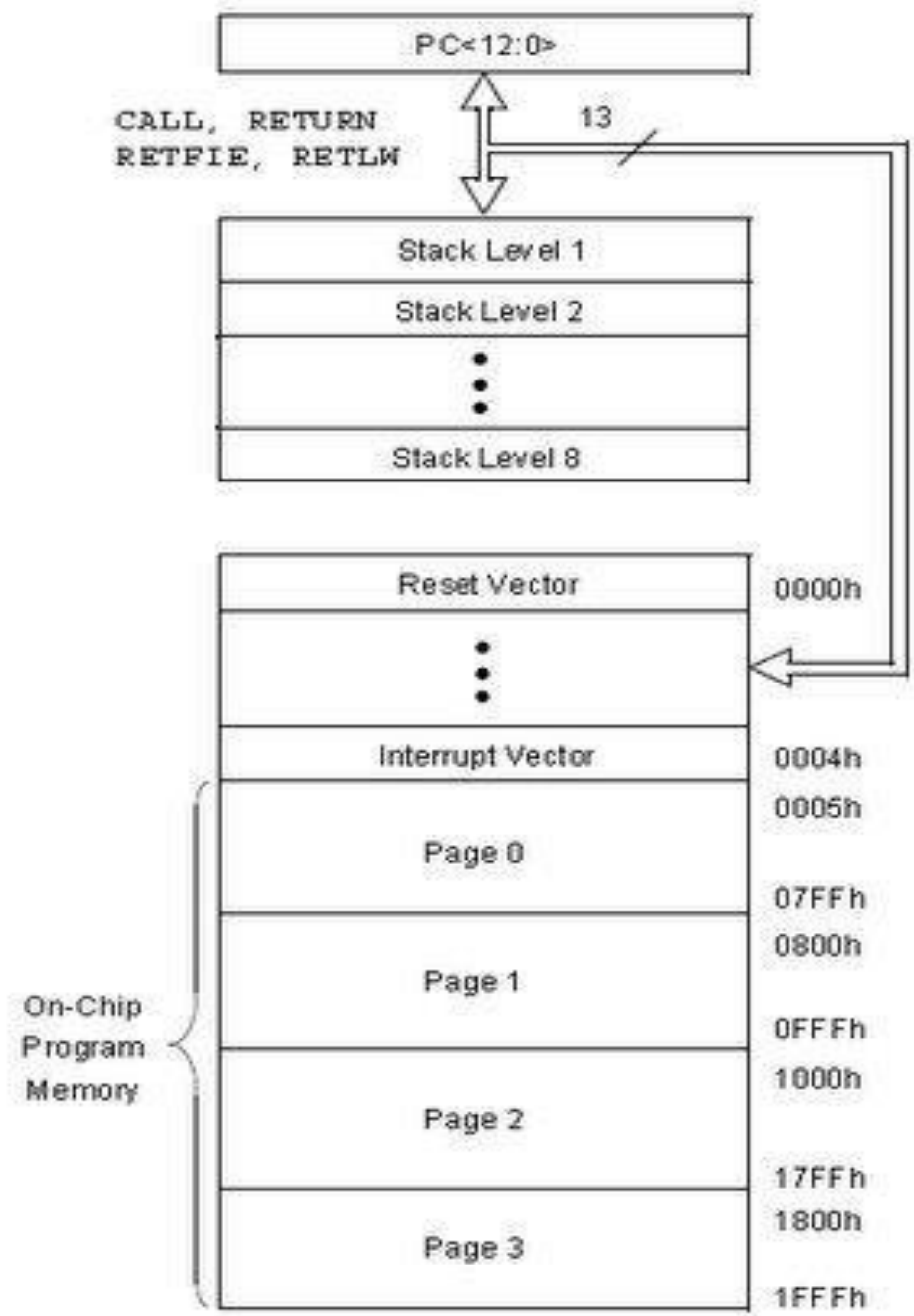
Memory of the PIC16F877..

- Each one of them has a different role. Program Memory and Data Memory two memories that are needed to build a program, and Data EEPROM is used to save data after the microcontroller is turn off.



PIC16F877A Program Memory

- Is Flash Memory
- Used for storing compiled code (user's program)
- Program Memory capacity is 8K x 14 bit → Each location is 14 bits long
  → Every instruction is coded as a 14 bit word
- PC can address up to 8K addresses
- Addresses H'000' and H'004' are treated in a special way



PIC16F877A Data Memory (RAM)

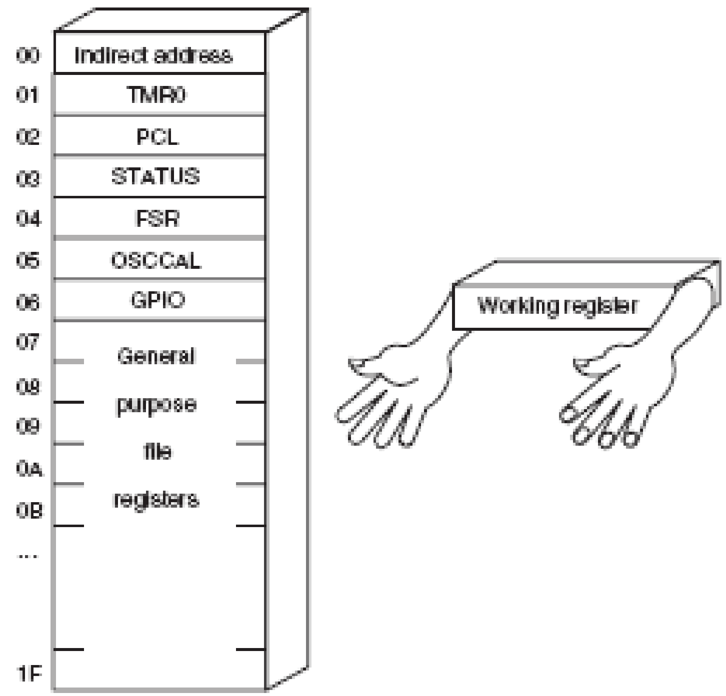
- Memory storage for variables
- Data Memory is also known as Register File and consists of two components.
  - General purpose register file (same as RAM).
  - Special purpose register file (similar to SFR in 8051).
- Addresses range from 0 to 511 and partitioned into 4 banks → each bank extends up to 7Fh (128 bytes).
- The user can only access a RAM byte in a set of 4 banks and only one bank at a time. **The default bank is BANK0.**
- To access a register that is located in another bank, one should access it inside the program. There are special registers which can be accessed from any bank, such as **STATUS** register.

PIC16F877A Registers

- Some CPU Registers:
  - W
  - PC
  - FSR
  - IDF
  - PCL
  - PCLATH
  - STATUS

W Register

- W, the working register, is used by many instructions as the source of an operand. This is similar to accumulator in 8051.
- It may also serve as the destination for the result of the instruction execution. It is an 8-bit register.



Program Counter

- Program Counter (PC) is 13 bit and capable of addressing an 8K word x 14 bit program memory space.
- PC keeps track of the program execution by holding the address of the current instruction.
- It is automatically incremented to the next instruction during the current instruction execution.

- **Program Counter Stack**

- an independent 8-level stack is used for the program counter.
- As the PC is 13-bit, the stack is organized as 8x13bit registers.
- When an interrupt occurs, the PC is pushed onto the stack. When the interrupt is being served, other interrupts remain disabled. Hence, other 7 registers of the stack can be used for subroutine calls within an **interrupt service routine** or within the mainline program.

FSR & INDF

- **FSR Register**

- (File Selection Register, address = 04H, 84H)

  is an 8-bit register used as data memory address pointer. This is used in indirect addressing mode.

- **INDF Register**

- (INDirect through FSR, address = 00H, 80H)

  INDF is not a physical register. Accessing INDF is actually access the location pointed to by FSR in indirect addressing mode.

PCL & PCLATH

- **PCL Register**
  - (**P**rogram **C**ounter **L**ow Byte, address =02H, 82H)

    PCL is actually the lower 8-bits of the 13-bit Program Counter. This is a both readable and writable register.

- **PCLATH Register**
  - (**P**rogram **C**ounter **LAT**c**H**, address = 0AH, 8AH)

    PCLATH is a 8-bit register which can be used to decide the upper 5-bits of the PC. PCLATH is not the upper 5bits of the PC. PCLATH can be read from or written to without affecting the PC. The upper 3 bits of PCLATH remain zero and they serve no purpose. When PCL is written to, the lower 5bits of PCLATH are automatically loaded to the upper 5bits of the PC.

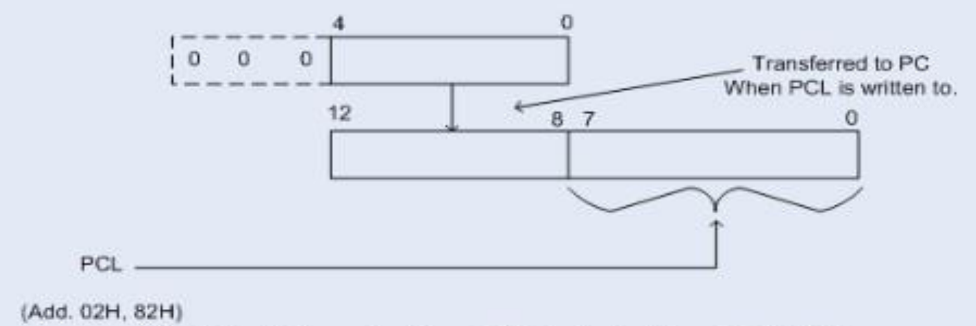


Fig 17.5 Schematic of how PCL is loaded from PCLATH

Memory Map Registers

- In order to start programming and build automated system, there is no need to study all the registers of the memory map, but only a few most important ones:
  - **STATUS register** – changes/moves from/between the
    - banks.
  - **PORT registers** – assigns logic values (“0”/”1”) to the ports
  - **TRIS registers** – data direction register (input/output)

STATUS Register

- Is an 8-bit register that stores the status of the processor.
- In most cases, this register is used to switch between the banks (Register Bank Select), but also has other capabilities.

STATUS REGISTER (ADDRESS 03h, 83h, 103h, 183h)

R/W-0	R/W-0	R/W-0	R-1	R-1	R/W-x	R/W-x	R/W-x
IRP	RP1	RP0	$\overline{TO}$	$\overline{PD}$	Z	DC	C
bit 7							bit 0

- **IRP - Register Bank Select bit.**
- **RP1:RP0: - Register Bank Select bits.**
- **TO: Time-out bit**
- **PD: Power-down bit**

Used in conjunction with PIC's sleep mode

- **Z: Zero bit**
- **DC: Digit carry/borrow bit**
- **C: Carry/borrow bit**

https://www.pcbheaven.com/picpages/The_Status_Register/

PIC16F877 Peripheral features

1. I/O Ports:

- PIC16F877 has **5 I/O ports**:
    - **PORT A** has 6 bit wide, Bidirectional
    - **PORT B, C, D** have 8 bit wide, Bidirectional
    - **PORT E** has 3 bit wide, Bidirectional
- In addition, they have the following alternate functions:

Port	Altenative uses of I/O pins	No.of I/O pins
Port A	A/D Converter inputs	6
Port B	External interrupt inputs	8
Port C	Serial port, Timer I/O	8
Port D	Parallel slave port	8
Port E	A/D Converter inputs	3
	Total I/O pins	33
	Total pins	40

PIC16F877 Peripheral features..

- Each port has **2 control registers**:
  - **TRISx** sets whether each pin is an input(1) or output(0)
  - **PORTx** sets their output bit levels or contain their input bit levels.
- Pin functionality “overloaded” with other features.
- Most pins have 25mA source/sink thus it can drive LEDs directly.

PIC16F877 Peripheral features…

2. Analog to Digital Converter (ADC)

- Only available in 14bit and 16bit cores
- Fs (sample rate) < 54KHz
- The result is a **10 bit digital** number
- Can generate an interrupt when ADC conversion is done
- The A/D module has **4 registers**:
  - A/D Result High Register (**ADRESH**)
  - A/D Result Low Register (**ADRESL**)
  - A/D Control Register0 (**ADCON0**)
  - A/D Control Register1 (**ADCON1**)
- Multiplexed **8 channel inputs**
  - Must wait $\mathbf{T}_{\mathbf{acq}}$ to change up sampling capacitor.
- Can take a reference voltage different from that of the controller.

PIC16F877 Peripheral features….

3. Timer/counter modules

- Generate interrupts on timer overflow
- Can use external pins as clock in/ clock out (ie. for counting events or using a different Fosc)
- There are **3 Timer/counter modules**:
  - Timer0: 8-bit timer/counter with 8-bit pre-scaler
  - Timer1: 16-bit timer/counter with 8-bit pre-scaler, can be incremented during SLEEP via external crystal/clock
  - Timer2: 8-bit timer/counter with 8-bit period register, pre-scaler and post-scaler.

PIC16F877 Peripheral features…..

4. Universal Synchronous Asynchronous Receiver Transmitter (USART/SCI) with 9-bit address detection.

- Asynchronous communication: UART (RS 232 serial)
  - Can do 300bps – 115kbps
  - 8 or 9 bits, parity, start and stop bits, etc.
  - Outputs 5V → needs a RS232 level converter (e.g MAX232)
- Synchronous communication: i.e with clock signal
  - SPI = Serial Peripheral Interface
    - 3 wire: Data in, Data out, Clock
    - Master/Slave (can have multiple masters)
    - Very high speed (1.6Mbps)
    - Full speed simultaneous send and receive (Full duplex)
  - I2C = Inter IC
    - 2 wire: Data and Clock
    - Master/Slave (Single master only)
    - Lots of cheap I2C chips available; typically < 100kbps

PIC16F877 Peripheral features…..

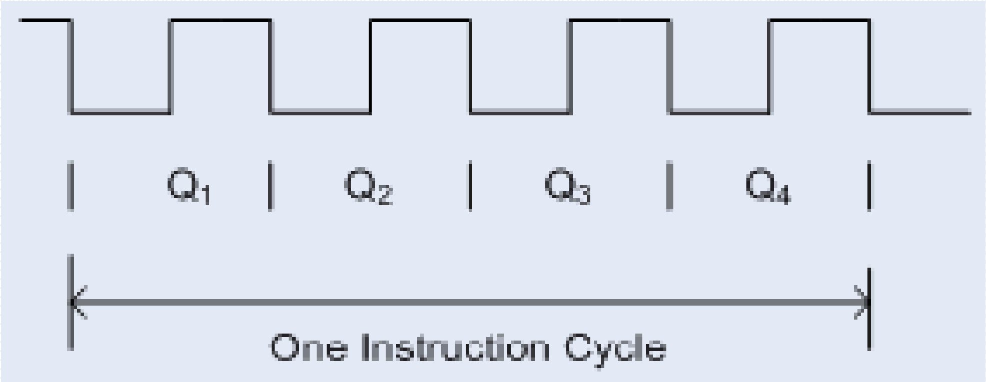
5. Capture, Compare, PWM modules

- Capture is 16-bit, max. resolution is 12.5 ns
- Compare is 16-bit, max. resolution is 200 ns
- PWM max. resolution is 10-bit

6. Parallel Slave Port (PSP) 8-bits wide, with external RD, WR and CS controls

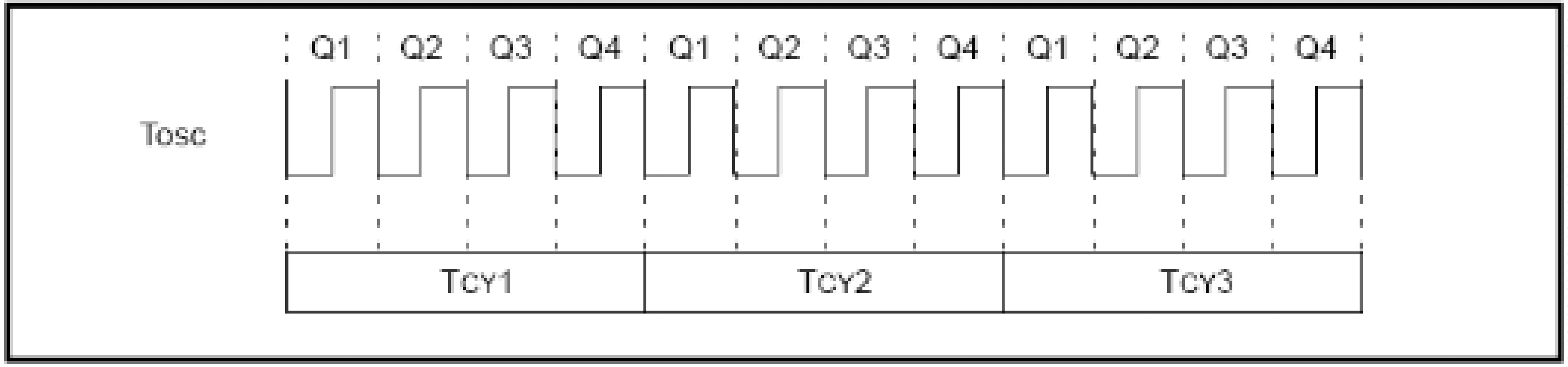
Clock and Instruction Cycles

- Clock from the oscillator enters a microcontroller via OSC1 pin where internal circuit of a microcontroller divides the clock into four even clocks Q1, Q2, Q3 and Q4 which do not overlap.
- These four clocks make up one **instruction cycle** (also called machine cycle) during which one instruction is executed.



Clock and Instruction Cycles..

- Execution of instruction starts by calling an instruction that is next in
- string.
- Instruction is called from program memory on every Q1 and is written in Instruction Register (IR) on Q4.
- Decoding and execution of instruction are done between the next Q1 and Q4 cycles. The following diagram shows the relationship between instruction cycle and clock of the oscillator (OSC1) as well as that of internal clocks Q1 – Q4.
- Program Counter (PC) holds information about the address of the
- next instruction.



Pipelining in PIC

- There are 35 single word instructions. A two-stage pipeline overlaps fetch and execution of instructions. As a result, all instructions execute in a single cycle except for program branches. These take two cycles since the fetch instruction is “flushed” from the pipeline while the new instruction is being fetched and then executed.
- A typical picture of the pipeline is shown in Figure 3.

Figure3: Instruction Pipeline Flow